Lab 1: Fine-tuning a Transformer for Patent Phrase Matching

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The task is to predict how similar an anchor and target phrase are, using the patent context. I used the DeBERTa-v3-small cross-encoder and Pearson correlation (r) to evaluate predictions. The training data contains 36,473 labelled rows and the competition test file contains 36 rows.

Task 1: Fine-tuned baseline

I used an anchor-grouped split: 28,914 training rows and 7,559 validation rows, with 147 complete anchors held out. The baseline validation Pearson score was **0.81601**, and training plus validation inference took about **5.9 minutes**. Holding out whole anchors is important because the same anchor can appear with many targets. A random row split could put the same anchor in both sets and make validation look better than performance on unseen anchors.

Task 2: Baseline without fine-tuning

Model	Validation Pearson
Frozen DeBERTa phrase encoder; cosine similarity (no fine-tuning)	0.19791
Fine-tuned DeBERTa-v3-small cross-encoder (Task 1)	0.81601

The frozen baseline encodes phrases separately and compares their vectors. The cross-encoder reads the phrase pair together, so its attention layers can learn token-level relationships during fine-tuning. The large difference suggests that task-specific training and joint phrase comparison are important for this dataset.

Task 3: Controlled experiments

I changed one setting at a time and kept the same validation split. The best result was obtained by removing the CPC section context, but the improvement was very small.

Run	Change	Val. r	Time	Interpretation
Baseline	Section context; LR $= 2 \times 10^{-5}$	0.81601	5.9 min	Reference
Experiment A	No context	**0.81675**	5.4 min	+0.00075; effectively similar
Experiment B	LR $= 1 \times 10^{-5}$	0.81326	5.8 min	Lower score in this run

The no-context run was selected because it had the highest validation score. However, the difference from baseline is much smaller than 0.005, so it may be random-seed noise. This notebook did not repeat these runs with another seed, so I cannot claim that removing context is reliably better.

Task 4: Error analysis

4.1 Pearson correlation by CPC section.

Section	A	B	C	D	E	F	G	H
Rows	943	1926	1177	199	373	968	1040	933
Pearson r	.785	.892	.772	.786	.806	.841	.710	.800

Section G (Physics) was the hardest, followed by C (Chemistry). Technical phrases in these areas may require specialist knowledge. Section B had the highest correlation, although the section results can be noisy, especially for D, which has only 199 rows.

4.2 Mean prediction by true score.

True score	0	0.25	0.5	0.75	1.0
Number of rows	1522	2465	2475	832	265
Mean prediction	0.101	0.291	0.479	0.642	1.032

Predictions for true scores of 0 and 0.75 move towards the middle: zeros average about 0.10, while 0.75 labels average about 0.64. Scores 0.25 and 0.5 are most common in the training set, so the model may be less accurate at the extremes. The mean prediction for score 1.0 is slightly above 1 because the model output is not clipped during validation.

4.3 Ten largest errors. The main pattern is high predictions for pairs labelled 0, such as *punch face / face punch* (0.846), *boom hydraulic cylinder / boom a hydraulic cylinder* (0.809), and *imaging axis / imaging of axis* (0.800). The model seems to rely on overlapping words. The opposite error also occurs: *breech mechanism / breech loading naval gun* has true score 0.75 but prediction 0.063, and *end pins / wood pin* has true score 0.75 but prediction 0.106. These pairs may need technical context or knowledge beyond surface wording. Some labels may look debatable, but these examples alone do not prove that the labels are wrong.

Task 5: Late submission and leaderboard scores

I selected the no-context configuration based on validation score before using the leaderboard results. The notebook generated a 36-row `submission.csv`. My only submission was Version 1, which Kaggle marked as successful after the deadline.

Model / version	Public score	Private score
DeBERTa-v3-small, Version 1	0.7924	0.8101

The validation Pearson score was 0.81675, compared with a public score of 0.7924 and a private score of 0.8101. The private score is closer to validation, while the public score is lower. These differences can happen because the leaderboard portions contain different examples and are limited in size. Validation performance can also be slightly optimistic after selecting the best configuration from experiments.

Submission	Validation Pearson	Public score	Private score
patent-phrase-lab1, Version 1	0.81675	0.7924	0.8101

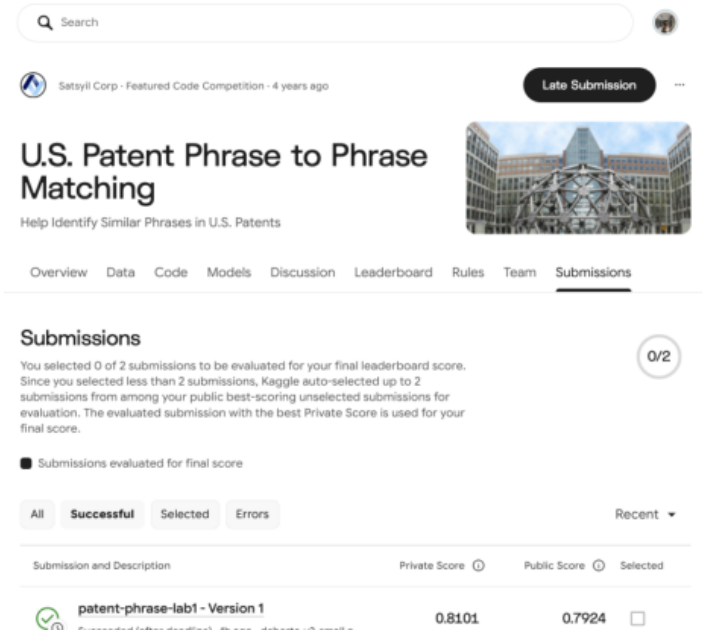


Figure 1: My Kaggle Submissions tab showing the successful Version 1 submission and its public/private scores.

Conclusion. Fine-tuning the cross-encoder performed much better than the frozen cosine-similarity baseline. Removing context produced the highest validation score, but the gain over the baseline was tiny and needs confirmation with another seed. Error analysis showed that word overlap can cause false high predictions, while related technical phrases with different wording can receive low predictions.

References: Kaggle, *U.S. Patent Phrase to Phrase Matching* competition; Aslanyan & Wetherbee (2022), *Patent Phrase Similarity Dataset*, arXiv:2208.01171; He et al. (2021), *DeBERTaV3*, arXiv:2111.09543; Reimers & Gurevych (2019), *Sentence-BERT*, arXiv:1908.10084; Hugging Face Transformers documentation; course starter notebook.